Comparing Text-Based Clinical Risk Prediction in Critical Care: A Note-Specific Hierarchical Network and Large Language Models

Jinghui Liu, Anthony Nguyen, Daniel Capurro, and Karin Verspoor

***Abstract*—Clinical predictive analysis is a crucial task with numerous applications and has been extensively studied using machine learning approaches. Clinical notes, a vital data source, have been employed to develop natural language processing (NLP) models for risk prediction in healthcare with robust performance. However, clinical notes vary considerably in text composition–written by diverse healthcare providers for different purposes–and the impact of these variations on NLP modeling is also underexplored. It also remains uncertain whether the recent Large Language Models (LLMs) with instruction-following capabilities can effectively handle the risk prediction task out-of-the-box, especially when using routinely collected clinical notes instead of polished text. We address these two important research questions in the context of in-hospital mortality prediction within the critical care setting. Specifically, we propose a supervised hierarchical network with note-specific modules to account for variations across different note categories, and provide a detailed comparison with strong supervised baselines and LLMs. We benchmark 34 instruction-following LLMs based on zero-shot, few-shot, and chain-of-thought prompting with diverse prompt templates. Our results demonstrate that the note-specific network delivers improved risk prediction performance compared to established supervised baselines from both measurement-based and text-based modeling. In contrast, LLMs consistently underperform on this critical task, despite their remarkable performances in other domains. This highlights important limitations and raises caution regarding the use of LLMs for risk assessment in the critical setting. Additionally, we show that the proposed model can be leveraged to select informative clinical notes to enhance the training of other models.**

***Index Terms*—Electronic health records, health information, large language models, natural language processing, predictive analysis.**

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I. Introduction

CLINICAL prediction models are important tools in healthcare to support clinical decision-making and have been studied for many decades [1], [2], [3], [4]. Risk stratification, a significant category of clinical predictive analysis, assesses the severity of patients based on available information, with wide implications for patient management [5], [6], [7], [8]. Traditionally, clinical risk prediction models are engineered by medical experts, who apply their medical knowledge and practical experience to create specific scoring systems [1]. In recent years, numerous investigations have focused on data-driven approaches to learn from real-world clinical data [9], [10], [11], demonstrating superior performance in clinical risk predictions. The interest in these machine learning methods has been further enhanced by the widespread implementation of electronic health record (EHR), increasing the availability of data for developing these models.

Although most clinical prediction models are based on structured data, such as clinical measurements, medical codes, or prescriptions, a significant amount of information is captured within clinical text written by various caregivers [12], [13], [14], such as nursing notes, radiology reports, and discharge summaries. Compared to structured data elements, unstructured clinical text contains more detailed information about patients, such as their history of illness and clinician's assessments, and can better reflect the complexity of patient conditions. Consequently, it has become increasingly common to apply Natural Language Processing (NLP) methods to train prediction models based on clinical text [14], [15], [16], leading to various applications in risk assessment [17], [18], early diagnosis [19], [20], and resource prediction [21], [22].

However, clinical text is written by different healthcare providers for various purposes, leading to significant variations in language and content [23]. For instance, nursing notes tend to focus on the patient's daily care and condition, while radiology reports are more diagnostic. These variations in clinical notes can significantly impact the performance of NLP models, as the models need to consider the nuances and context of the text to make accurate predictions. Meanwhile, most existing NLP models do not account for these variations, and it remains unclear how they affect the models' performance.

Furthermore, a paradigm shift in NLP research has been the development of Large Language Models (LLMs) that can

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perform well on NLP tasks by simply being instructed or prompted [24], [25], [26]. Pretrained LLMs demonstrate impressive capabilities in performing tasks in zero-shot setting without being further fine-tuned, which has drawn significant attention from the healthcare community [27], [28], [29]. Recent studies have shown that LLMs can achieve superior performance on medical question answering [30], [31], most famously on USMLE exams [32], and can handle various diagnostic tasks in different clinical specialties [33], [34], [35]. However, many of these studies examined LLMs using curated datasets, such as multiple-choice question answering, rather than routinely collected data from real-world practice. The performance of LLMs for clinical predictive analysis in this setting, especially in comparison with classical supervised ML and NLP models, is less explored.

In this study, we consider two main research questions: *1) Can considering the variations in clinical notes further enhance the performance in text-based clinical risk prediction? 2) How do supervised text-based models compare with instruction-following LLMs for clinical risk prediction in the critical care setting using real-world clinical text?* To address these questions, we conduct a benchmarking study to compare the performance of **prompt-based LLMs** with **supervised text-based models**, and propose a novel note-specific hierarchical model that accounts for the variations in clinical notes. We evaluate the models on a canonical in-hospital mortality prediction task in Intensive Care Units (ICUs), which is widely studied in the literature using both routinely collected structured measurements [6], [36] and unstructured text [18], [37], [38]. Our contributions are as follows:

- We conduct **benchmarking of 34 popular LLMs on in-hospital mortality prediction**, exploring model sizes ranging from 0.5B to 72B parameters and employing various prompting and decoding strategies. Our results reveal several LLMs that produce competitive risk assessment scores based on routinely collected clinical text, although the best LLM still lags behind supervised text-based baselines.
- To enhance predictive accuracy, we propose **a supervised note-specific hierarchical model** that accounts for variations in clinical note categories. Our results show that the proposed model outperforms previous supervised text-based models–including fine-tuned Transformer encoders and decoders–and is even competitive with a strong measurement-based baseline.
- We also demonstrate that the proposed model can **identify important clinical notes** using an model explanation method known as attention flow [39]. By selecting the most informative notes through attention flow, the model can significantly benefit systems that rely on shorter input contexts.

II. Related Works

Clinical predictive analysis, such as mortality prediction, is widely studied in the literature [10], [40], [41], [42]. Clinical notes have been shown as a valuable source of input in the development of these prediction models [16], [43], and also provide potential benefits of surfacing human documentation to support informed actions in the modeling process [44], [45]. Meanwhile, modeling clinical notes often underperforms that based on structured clinical measurements, such as vital signs and laboratory results, in predicting patient risk [46], [47]. Top measurement-based models, especially those can manage missing values, still provide most accurate results in clinical prediction tasks, although in comparison to text-based models they could be less robust and suffer more from data distribution shifts [48], [49].

LLMs have been widely studied in healthcare since the introduction of ChatGPT, which has been shown effective in answering complex medical questions [30], [32], [50]. They have been analyzed across a range of clinical specialties, such as cardiology [51] and hepatology [52], but mostly focusing on question answering. Furthermore, these studies tend to use curated datasets with relative short input text. In contrast, text-based predictive modeling often requires processing an aggregation of diverse and long clinical notes, which can be more challenging for LLMs. A recent study proposed a long clinical document benchmark on mortality prediction [18], which examines various methods including prompt-based LLMs to perform 30-day mortality prediction based on hospital discharge summaries. They prompt 7 LLMs to produce a binary class in a single word (0: alive or 1: dead) and report GPT-4 achieving the highest, yet still modest, F1 score of 32.2%. Our work differs from this previous study in four key ways. First, we focus on in-hospital mortality prediction from the first day of admission to the ICU, a more challenging task due to rapidly evolving patient conditions during the hospital stay. In addition, the input notes are more diverse and noisier compared to a single type of discharge summary. Second, we ask the LLMs to produce a risk assessment score (0-100) instead of a binary label (0 or 1), which is critical for stratification and triage in practice. This also enables evaluation using more robust metrics including AUR-ROC and AUC-PRC. Third, we examine a wider range of LLMs, prompts, and decoding methods, exploring their impact on the task. Although proprietary models are excluded due to restrictions from the relevant data usage agreement associated with the MIMIC-III dataset we use [53], we include the latest open-source LLMs as of early 2025 in the experiments. Fourth, we compare the LLMs with both text-based and measurement-based models to provide a more comprehensive benchmarking study.

Clinical text is known to contain subdomains and notes written by different clinicians and can present distinct linguistic features and medical concepts [23], [54]. Meanwhile, the value of considering these variations in designing NLP models is underexplored. We are inspired by the modular learning approach in language models [55], [56] and propose a model that takes into account the variations in clinical notes by note-specific modules. Our model is similar to the Mixture of Experts (MoE) architecture, which has been shown to be effective in scaling up LLMs [57], [58]. We simplify this idea by using metadata to route notes to the corresponding modules, unlike most MoE models that contain a learned router to assign modules or experts.

TABLE I
DEMOGRAPHICS OF THE EVALUATION COHORT FOR IN-HOSPITAL MORTALITY PREDICTION

	Value
Number of patients	3,386
Age (mean±std)	64.0±17.3
Female	1487 (43.9%)
Mortality (in-hospital)	391 (11.5%)
Ethnicity	
Caucasian	2406 (71.1%)
African American	376 (11.1%)
Hispanic	85 (2.5%)
Asian	68 (2.0%)
Unknown/others	451 (13.3%)
Admission type	
Emergency	2872 (84.8%)
Elective	423 (12.5%)
Urgent	91 (2.7%)

III. MATERIALS AND METHODS

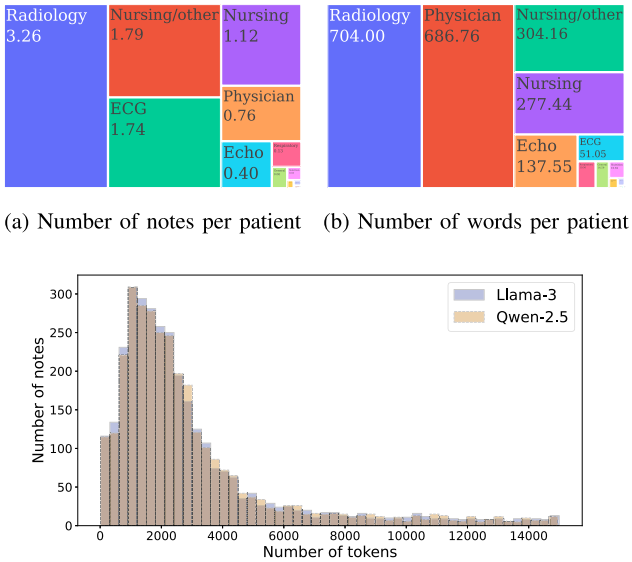
A. Dataset and Task

We utilize the Medical Information Mart for Intensive Care (MIMIC) III version as the data source for our experiments [53], which is a public EHR database collected from Beth Israel Deaconess Medical Center. We use MIMIC-III rather than MIMIC-IV because the latter has not released progress notes written during the hospital stay as of the time of our study. We follow [59] to prepare a cohort of patients for in-hospital mortality prediction by the first day of ICU admission, which includes patients over age of 15 and has only one ICU encounter during the hospital episode. We use their test set of 3,386 admissions to report model performances on. The patient demographics and in-hospital mortality rate of this evaluation cohort are shown in Table I. LLMs are prompted to assess these 3,386 patients without any parameter updates, while the supervised models are trained on the training set of 26,754 patients and validated on 3,375 patients following the same study [59].

For text-based modeling, all clinical notes charted prior to the 24th hour of admission are extracted, sorted by their chart time, and used as model input. Patients on average have 9.3 notes with a standard deviation of 6.2, and the 90 percentile is 18 notes. Each patient would have numerous notes from different categories, such as nursing notes and radiology reports, which are aggregated into a time series. Fig. 1(a) and (b) show the numbers of notes and words per patient by note categories. Each patient has on average 2112 words in their notes, with 90 percentile being 5194 words. These numbers further increased after tokenization when using LLMs. For example, Llama-3 tokenizer would increase the mean and 90 percentile to 4104 and 10665 tokens, respectively. As an illustration, the distributions of tokens by two different LLM tokenizers are shown in Fig. 1(c).

For measurement-based modeling, we follow the pipeline from [60], a high-quality benchmark on MIMIC-III, to extract and preprocess 104 curated clinical measurements, including vital signs and laboratory results. The pipeline also includes handling missing values and outliers by imputation and clipping. We refer readers to the original paper for a complete list of measurements.

We calculate the Area Under the Receiver Operating Characteristic Curve (AUC-ROC), the Area Under the Precision-Recall Curve (AUC-PRC), and F1 score to evaluate the model performance. We compare the performance of the LLMs with the supervised text-based models and the measurement-based models, all evaluated on the same test set.



(a) Number of notes per patient (b) Number of words per patient

(c) Distributions of tokens per patient by two different LLM tokenizers. Similar to [18], we exclude the numbers over 15,000.

Fig. 1. Distributions of clinical note types and input lengths.

B. Prompting Large Language Models for Risk Prediction

We prompt 34 popular open-source, instruction-tuned LLMs for the in-hospital mortality prediction task based on an aggregation of the clinical notes. These LLMs cover a wide range of model sizes and versions, including the Llama series from Meta [61], [62], Mistral and Mixtral from MistralAI [58], [63], the Qwen series from Alibaba [64], [65], the Phi series from Microsoft [66], the Gemma series from Google [67], and five medically adapted models [68], [69], [70], with parameters ranging from 0.5B to 72B.

We create the prompt template by asking ChatGPT (4o) to improve a seeding instruction, by which we create a system prompt and a user instruction to ask LLMs to provide a risk assessment score between 0 and 100. Fig. 2 shows an illustration of such a template. In practice, the system and user prompts are customized to the specific chat template of each LLM by the corresponding tokenizer. We explore five different user instructions and explore how sensitive the LLM performance is to the prompt variations, and report mean and standard deviation of the risk scores from the five prompts.

We apply this prompting strategy to all LLMs and record the risk scores for the 3,386 patients in the evaluation cohort. For top-performing LLMs, we further consider Chain-of-Thoughts (CoT) [71] and few-shot prompting strategies, which are usually beneficial for general-domain NLP tasks. For CoT prompting, we follow previous work [72] by adding "Let's think step by step" before asking for risk scores. We further apply self-consistency (SC) [73] for CoT to improve the model

Prompt Template

[SYSTEM PROMPT] You are a medical artificial intelligence assistant designed to provide accurate, evidence-based, and professional assessments of patient health concerns. In addition to offering clear and empathetic explanations, you are tasked with calculating and delivering an assessment score that reflects the patient's risk level based on the provided information. Ensure that all risk scores are grounded in current medical guidelines and best practices. Emphasize that these scores are informational and should not replace personalized medical advice from a healthcare professional. Always prioritize patient safety, confidentiality, and encourage follow-up with a qualified provider.
[USER PROMPT] Based on the clinical notes provided by the healthcare team, calculate a numerical risk score for the patient's in-hospital mortality. The score should range from 0 to 100, where higher numbers indicate greater mortality risk. Provide only the numerical score, without additional commentary. The patient's clinical notes are as follows:

[aggregated clinical notes]

Risk score:

Fig. 2. Prompt template for risk assessment with LLMs.

performance. For few-shot prompting, we provide four examples randomly selected from the train set in [59], with two positive and two negative examples. We also gather five variations by changing the random seed for sampling the few-shot examples.

We decode the LLM outputs using nucleus sampling by setting the top-p probability to 0.9 and temperature to 0.8. We additionally perform greedy decoding, and report the result in place of nucleus sampling if the final mean AUC-ROC is higher. We limit the maximum number of new tokens to 50 to avoid excessive computation, but increase that to 250 for CoT prompting to allow additional reasoning. Greedy decoding was not applied to CoT with self-consistency. For postprocessing, we remove any regex matching that describe patient age (e.g., *xx-year-old*) and then retrieve the number from 0 to 100 as the risk score. If no valid score is found, we assign a score of 50 to the patient and count the case as missing.

Alternatively, we also prompt LLMs to perform binary classification by generating a target label, i.e., survived or deceased. Compared to generating a risk score, it is expected that LLMs are more familiar with generating a token given how they are pretrained. We follow [18] to update the user prompt in Fig. 2 as *"How likely is the given patient's in-hospital mortality? Please only answer with one word: 0:alive, 1:dead. The patient's clinical notes are as follows:"*. We prompt the LLMs five times using this same prompt and report averaged results. However, we do not consider it as our main experimentation given that numerical values are essential in practice to stratify patients [1], [74], [75], which could not be achieved with binary outputs.

All inferences were performed using the vllm library (0.6.6.post1) on NVIDIA H100 GPUs. Models with sizes below 20B were inferenced in 16-bit precision, while larger models were quantized to 4-bit precision using built-in bitsandbytes quantization provided in vllm.

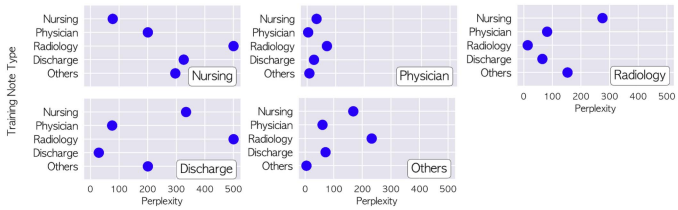


Fig. 3. Perplexity scores of language models (GPT-2) trained and evaluated on different note categories.

C. An Inter-Corpus Analysis: Variations in Note Categories

In this section, we conduct a case analysis to illustrate the variations in different clinical note categories, which motivates our proposed note-specific model in Section III-D. We sample texts from four most common note categories (i.e., nursing notes, physician notes, radiology reports, and discharge summaries) and other infrequent note categories (grouped as "Others"), and construct note-specific corpus for each of them. We then perform language modeling on one category of notes and evaluate on the rest, following previous works in calculating inter-corpus similarity and intra-corpus homogeneity using LMs [76], [77], [78]. We report the perplexity scores for language modeling, where lower perplexity means the similarity between training and testing corpora that represent two clinical note categories.

The results in Fig. 3 reveal clear differences in the language modeling performance across different note categories, with the lowest perplexity scores being achieved when training and testing occur on the same category. When applied to unseen text, there are also distinctions between the different categories. These variations demonstrate the need of considering the note-specific characteristics in the model design.

D. The Proposal: A Note-Specific Network

Towards this purpose, we propose a Note-Specific Hierarchical Network (NSHNET) with feedforward modules that handle specific note categories, illustrated in Fig. 4. The model consists of an embedding module, a note encoder, a patient encoder, and an MLP prediction head. The patient encoder is a Transformer-encoder model, similar to BERT [79], but with different feedforward networks (FFN) that are dedicated to each category of clinical notes. For example, *nursing notes* would only be processed by *Nursing FFN*, similarly *radiology notes* by *Radiology FFN*. This resembles the MoE system, where different parts of the input are routed to and handled by specialized networks, except that the routing in our patient encoder is based on the metadata, where the model would switch modules according to the note types. A [CLS] token is appended to the note sequence, which interacts with other note tokens through attention and is used as the final patient representation. We also explored alternative token pooling strategy with additional attention layer to obtain the patient representation, but found similar performance.

As a supervised classifier, we train NSHNET on the training set with a learning rate of 5e-4 under a cosine schedule, and

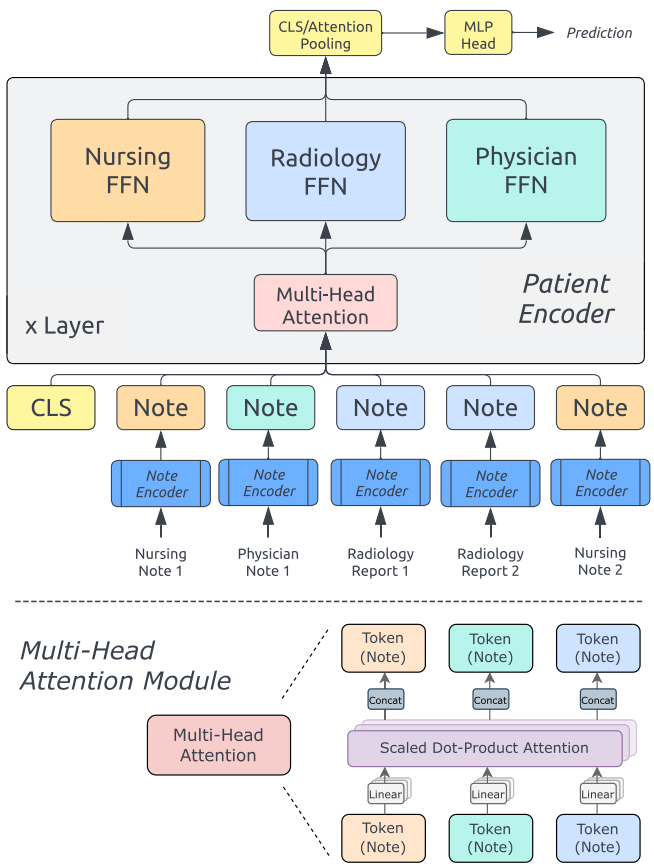


Fig. 4. We examine a hierarchical, Transformer-based model with note-specific modules in the patient encoder to handle varied types of clinical notes for predictive analysis, denoted as NSHNET. The clinical notes are first encoded as *note tokens*, which are then processed by the *multi-head attention module*, followed by *note-specific feedforward layers* (FFN).

we perform early stopping based on the validation set. Given previous observations on the instability of Transformers training [80], [81], we opt not to train a stack of Transformers from scratch on this a relatively small, task-specific dataset. Instead, we adopt a much simpler convolutional neural network (CNN) as note encoder to improve training efficiency and stability, which has 128 CNN kernels with widths of 1, 3, and 5 tokens. The patient encoder considers four categories of notes according to the proportions in Fig. 1, including radiology, physician, and nursing notes, plus a combined category of other notes. Discharge summaries are excluded since the model is applied during the early hospital admission. To stabilize training, we include only a single layer of Transformer in the patient encoder as we observed similar performance with 2 or 4 layers in preliminary experiments, which is likely limited by the weak capabilities of the note encoder. Although increasing the complexity of both the note encoder and dataset size may yield better performance, we leave further hyperparameter exploration to future work.

E. Supervised Baseline Models

We consider a range of representative supervised baseline methods as comparison. The measurement-based baseline is GRU-D [82]–a strong model designed to model multivariate time-series data in EHR–trained and applied on the 104 measurements from MIMIC-Extract [60]. Supervised text-based models include non-Transformer baselines from [59], which are Hierarchical Attention Network (HAN) [83], [84] and MultiResCNN [85]).

Furthermore, we fine-tune various encoder-based pretrained language models, including the classic BERT model [79] with its clinical and biomedical variants: ClinicalBERT [86] and PubMedBERT [87], based on the corresponding checkpoints. We also include two efficient encoder models–Longformer [88] and BigBird [89]–to handle long input, for which we increase the maximum input length from 512 (for BERT) to 2048 tokens and initialize using the clinically adapted checkpoints from [90].

Finally, we also fine-tune small-sized LLMs using low-rank adaptation (LoRA) [91], which has become a standardized practice in efficiently customizing LLMs. We include four representative models: LLAMA-3.1-8B, QWEN-2.5-7B, MISTRAL-0.3-7B, and PHI-MINI. We set the LoRA rank to 16, alpha to 32, dropout to 0.05, and attach LoRA modules to all linear layers [92], with 8 b quantization using bitsandbytes. Different from fine-tuning these decoder-based models for generation, a classification layer is attached to process the final token to perform the prediction task. All pretrained language models (BERT, efficient encoders, and decoder-based LLMs) are fine-tuned using a learning rate of 2e-5 with 500 steps for linear warm-up and with early stopping.

F. Identifying Important Notes Based on Attention Flow

Multiple previous works have observed that different types of clinical note have various levels of predictive value [46], [59], [93]. A recent study explores how to select the most informative notes to support other models in the clinical setting [94], which is an important application since a patient can have many notes charted, resulting in tens of thousands of words that cannot be easily handled by many current models. For example, BERT [79] has been used as the basis for multiple clinical prediction tasks in a multi-site study [95], but the input capacity of 512 tokens in BERT means that deliberate selections on the input need to be made when using the pretrained checkpoints. This can be challenging even for LLMs with long context windows [96].

We explore using NSHNET to identify notes with most contributions by calculating the attention flow in the patient encoder. Attention flow is a post hoc method proposed to interpret Transformer encoders [39], which is shown equivalent to Shapley Values to attribute the influence of tokens on a given one [97], making it a more effective mechanism to capture model confidence and to provide interpretability compared to raw attention. We calculate the flow values of different notes with respect to the [CLS] token that is used as the final representation. We then examine the predictive values of the identified notes by fine-tuning BERT models for the same risk prediction task using these notes. We sort the notes according to the attention flow values, concatenate all the notes, and feed the first 512 tokens to the BERT models, which is compared with the default practice to concatenate notes in temporal order.

TABLE II
RESULTS ON IN-HOSPITAL MORTALITY PREDICTION AT THE 24TH HOUR OF ICU ADMISSION

	AUC-ROC	AUC-PRC	F1	Missing rate
Prompt-based LLMs for Zero-shot Prediction				
Small-sized LLMs (<20B)				
Llama-2-7b-chat-hf	0.571 (0.026)	0.136 (0.014)	0.116 (0.028)	34.9% (0.5%)
Llama-2-13b-chat-hf	0.585 (0.044)	0.191 (0.029)	0.189 (0.030)	34.8% (0.3%)
Llama-3-8B-Instruct	0.627 (0.024)	0.275 (0.030)	0.202 (0.004)	13.1% (0.0%)
Llama-3.1-8B-Instruct	0.576 (0.021)	0.175 (0.017)	0.222 (0.009)	57.1% (16.4%)
Llama-3.2-1B-Instruct	0.570 (0.023)	0.144 (0.012)	0.226 (0.014)	1.3% (1.0%)
Llama-3.2-3B-Instruct	0.524 (0.024)	0.136 (0.014)	0.190 (0.012)	16.9% (7.5%)
Qwen2-7B-Instruct	0.688 (0.008)	0.257 (0.024)	0.222 (0.003)	3.5% (1.0%)
Qwen2.5-7B-Instruct	0.610 (0.016)	0.166 (0.011)	0.235 (0.013)	30.3% (2.3%)
Qwen2.5-14B-Instruct	0.583 (0.019)	0.191 (0.027)	0.185 (0.014)	61.1% (7.0%)
Mistral-7B-Instruct-v0.2	0.588 (0.020)	0.196 (0.018)	0.201 (0.035)	75.0% (4.7%)
Mistral-7B-Instruct-v0.3	0.531 (0.006)	0.150 (0.014)	0.126 (0.035)	83.6% (3.4%)
Phi-3-mini-128k-instruct	0.525 (0.003)	0.152 (0.024)	0.017 (0.005)	81.9% (1.6%)
Phi-3-medium-128k-instruct	0.730 (0.026)	0.321 (0.067)	0.263 (0.007)	10.1% (18.3%)
Llama3-OpenBioLLM-8B	0.553 (0.009)	0.168 (0.017)	0.214 (0.007)	21.9% (6.2%)
Medalpaca-7b	0.496 (0.011)	0.101 (0.005)	0.074 (0.008)	71.0% (0.2%)
Meditron-7b	0.518 (0.019)	0.121 (0.006)	0.085 (0.008)	71.7% (0.7%)
Llama-3-8B-UltraMedical	0.525 (0.004)	0.193 (0.023)	0.165 (0.022)	76.9% (8.8%)
Large-sized LLMs (>20B)				
Llama-2-70b-chat-hf	0.618 (0.029)	0.182 (0.024)	0.196 (0.026)	40.0% (6.1%)
Llama-3-70B-Instruct	0.741 (0.008)	0.371 (0.023)	0.245 (0.006)	13.1% (0.0%)
Llama-3.1-70B-Instruct	0.764 (0.014)	0.363 (0.035)	0.231 (0.007)	0.1% (0.2%)
Llama-3.3-70B-Instruct	0.781 (0.006)	0.403 (0.051)	0.226 (0.006)	0.0% (0.0%)
Qwen2-72B-Instruct	0.758 (0.006)	0.432 (0.010)	0.229 (0.009)	2.3% (0.9%)
Qwen2.5-72B-Instruct	0.683 (0.007)	0.290 (0.019)	0.248 (0.002)	5.9% (1.7%)
Mixtral-8x7B-Instruct-v0.1	0.613 (0.027)	0.217 (0.019)	0.235 (0.031)	63.6% (6.6%)
Gemma-2-27b-it	0.541 (0.028)	0.165 (0.025)	0.175 (0.015)	60.3% (9.3%)
Llama-3-70B-UltraMedical	0.518 (0.010)	0.122 (0.010)	0.120 (0.024)	70.0% (8.6%)
Supervised Measurement-based Model				
GRU-D (104 clinical measurements)	**0.887**	**0.546**	0.388	N/A
Supervised Text-based Models				
Non-Transformer baselines				
Hierarchical Attention Network	0.856 (0.004)	0.456 (0.017)	-	N/A
MultiResCNN	0.866 (0.002)	0.484 (0.007)	-	N/A
Pretrained encoder-based language models				
BERT	0.774 (0.009)	0.306 (0.024)	0.153 (0.078)	N/A
ClinicalBERT	0.788 (0.004)	0.347 (0.017)	0.304 (0.048)	N/A
PubMedBERT	0.816 (0.002)	0.392 (0.007)	0.344 (0.006)	N/A
ClinicalLongformer	0.840 (0.006)	0.439 (0.008)	0.421 (0.019)	N/A
ClinicalBigBird	0.842 (0.001)	0.451 (0.010)	0.397 (0.011)	N/A
Decoder-based LLMs fine-tuned with LoRA				
Llama-3.1-8B (LoRA)	0.884 (0.000)	0.530 (0.002)	0.415 (0.004)	N/A
Qwen-2.5-7B (LoRA)	0.878 (0.003)	0.525 (0.004)	0.401 (0.016)	N/A
Mistral-7B-v0.3 (LoRA)	0.875 (0.001)	0.510 (0.007)	0.422 (0.001)	N/A
Phi-3-mini (LoRA)	0.860 (0.001)	0.460 (0.019)	0.339 (0.008)	N/A
Proposed				
NSHNET	0.882 (0.002)	**0.546 (0.005)**	**0.481 (0.026)**	N/A

Results are presented in mean and standard deviation for LLMs (averaged over five different prompts) and the supervised text-based models (over five random seeds). We use the default 0.5 as the classification threshold when calculating F1.

IV. RESULTS

Table II shows the mortality prediction performance of zero-shot prompted LLMs, supervised measurement-based model, and supervised text-based models. We consider AUC-ROC scores over 0.6, AUC-PRC and F1 over 0.2 as acceptable performance and color them in green. We also consider the missing rate for the LLM, which is the proportion of patients that the model did not output any valid score. We consider a missing rate below 10% as acceptable, also colored in green.

A. Performance of Prompt-Based LLM

The majority of the 25 small-sized LLMs [1] with parameters below 20B perform poorly on the task, with only four models achieving acceptable performance on AUC-ROC. Among these, PHI-3-MEDIUM-128K-INSTRUCT shows the best performance, with AUC-ROC at 0.73 and AUC-PRC over 0.3, but with a missing rate bordering 10%. While some small-sized LLMs may have low missing rates, such as LLAMA-3.2-1B-INSTRUCT, they could also show poor performance. In addition, newer models from the same model family are not necessarily better than their predecessors. For example, LLAMA-3-8B-INSTRUCT shows better performance than LLAMA-3.1-8B-INSTRUCT and LLAMA-3.2-3B-INSTRUCT.

On the other hand, the large-sized LLMs show significantly better performances, with most models achieving acceptable AUC-ROC, and four models achieving both high performances and very low missing rates. Among them, LLAMA-3.3-70B-INSTRUCT and QWEN2-72B-INSTRUCT show the best overall performance, with AUC-PRC over 0.75 and F1 over 0.4. Comparing QWEN2-72B-INSTRUCT with its smaller counterpart QWEN2-7B-INSTRUCT, the results illustrate the significant benefits of scaling up the model size for this task.

We consider five LLMs that perform well in the zero-shot setting to examine alternative prompting strategies. Table III

[1]We excluded 8 small-sized models with NaN outputs from Table II: Phi-4, Gemma-1.1 (2B and 7B), Gemma-2 (2B and 9B), Qwen-2.5 (0.5B, 1.5B, 3B).

TABLE III
LLM RESULTS ON WITH ADDITIONAL PROMPTING STRATEGIES

	AUC-ROC	AUC-PRC	F1
Qwen2-7B-Instruct	0.687	0.257	0.222
+CoT	0.643 (6.4%↓)	0.208 (19.1%↓)	0.222 (0.0%)
+CoT+SC	0.638 (7.1%↓)	0.220 (14.4%↓)	0.226 (1.8%↑)
+CoT+SC (Mean)	0.658 (4.2%↓)	0.218 (15.2%↓)	0.222 (0.0%)
Phi-3-medium-128k-instruct	0.722	0.319	0.263
+CoT	0.665 (7.9%↓)	0.215 (32.6%↓)	0.251 (4.6%↓)
+CoT+SC	0.659 (8.7%↓)	0.245 (23.2%↓)	0.270 (2.7%↑)
+CoT+SC (Mean)	0.699 (3.2%↓)	0.251 (21.3%↓)	0.258 (1.9%↓)
Llama-3.1-70B-Instruct	0.764	0.363	0.231
+CoT	0.717 (6.2%↓)	0.243 (33.1%↓)	0.223 (3.5%↓)
+CoT+SC	0.734 (3.9%↓)	0.249 (31.4%↓)	0.234 (1.3%↑)
+CoT+SC (Mean)	0.764 (0.0%)	0.294 (19.0%↓)	0.220 (4.8%↓)
Llama-3.3-70B-Instruct	0.781	0.403	0.226
+CoT	0.738 (5.5%↓)	0.293 (27.3%↓)	0.221 (2.2%↓)
+CoT+SC	0.745 (4.6%↓)	0.309 (23.3%↓)	0.224 (0.9%↓)
+CoT+SC (Mean)	0.774 (0.9%↓)	0.325 (19.4%↓)	0.219 (3.1%↓)
Qwen2-72B-Instruct	0.757	0.432	0.229
+CoT	0.726 (4.1%↓)	0.324 (25.0%↓)	0.226 (1.3%↓)
+CoT+SC	0.723 (4.5%↓)	0.369 (14.6%↓)	0.232 (1.3%↑)
+CoT+SC (Mean)	0.752 (0.7%↓)	0.354 (18.1%↓)	0.225 (1.7%↓)

All results used nucleus sampling for decoding.

TABLE IV
RESULTS OF DIFFERENT BERT VARIANTS ON MORTALITY PREDICTION WITH OR WITHOUT NOTE SELECTION BY NSHNET

	AUC-ROC	AUC-PRC	F1
BERT	0.786	0.337	0.231
BERT + selection	0.848 (7.9%↑)	0.444 (31.8%↑)	0.330 (42.9%↑)
ClinicalBERT	0.787	0.344	0.286
ClinicalBERT + selection	0.855 (8.6%↑)	0.487 (41.6%↑)	0.418 (46.2%↑)
PubMedBERT	0.812	0.364	0.290
PubMedBERT + selection	0.854 (5.2%↑)	0.490 (34.6%↑)	0.422 (45.5%↑)

presents the baseline results of zero-shot prompting with nucleus sampling, and results with additional CoT, SC, and a self-consistency variation to replace major voting to averaging, denoted as *SC (Mean)*. We find that model performances drop significantly with the addition of CoT, with standard SC only occasionally recovering some of the performance. SC with averaging shows much better improvements over CoT, but still falls short of the original performance. Adding few-shot demonstrations further degrades the performance of the five models, mostly reducing AUC-ROC lower than 0.6, so we do not include them in the table. These findings suggest the unique challenges of prompt-based learning to handle long clinical notes for patient risk prediction.

We additionally prompt three best performing LLMs by generating classification tokens instead of risk scores. This helps to increase the F1 scores of LLAMA-3.3-70B-INSTRUCT from 0.226 (0.006) to 0.271 (0.003), QWEN2-72B-INSTRUCT from 0.248 (0.002) to 0.281 (0.006), and PHI-3-MEDIUM-128K-INSTRUCT from 0.263 (0.007) to 0.293 (0.006), while we cannot compute AUC scores given the output format. This limit the practical utility of these methods in real-world where it is essential to leverage risk scores to adjust the threshold for sensitivity or for appropriate alerts [74], [75]. Furthermore, they still fall short compared to supervised results despite the increase.

B. Supervised Results and NSHNET

The supervised baselines still represent the state-of-the-art in this task and mostly outperform the prompt-based LLMs (Table II). The BERT models show poor performances given their limited input size (512 tokens) compared to other supervised text models, with their AUC-PRC scores lower than LLAMA-3.3-70B-INSTRUCT and QWEN-2-72B-INSTRUCT. Meanwhile, the supervised results improved by a large margin with long input context, where even the non-Transformer baselines–HAN and MultiResCNN–show better performance than even the best LLM.

While encoder-based pretrained models show improved but still limited supervised performances, we observe strong results by fine-tuning decoder-based LLMs using LoRA, with the best model–LLAMA-3.1-8B (LORA)–approaching the supervised measurement-based model in AUC-ROC. This shows the capacities of modern LLMs to be adapted for various modeling settings. Compared to their zero-shot counterparts, all fine-tuned models show substantial improvements. Meanwhile, we also note that due to their size, even parameter-efficient fine-tuning requires a significant amount of computation compared to other supervised methods.

The proposed NSHNET obtains similar or better results compared to all the text-based baselines. Specifically, it achieves similar AUC-ROC as the much larger LLAMA-3.1-8B (LORA) and higher AUC-PRC and F1 than other text models. We consider this can be attributed to NSHNET's capability to focus on important notes and model them with dedicated modules, thus filtering out noisy content to achieve strong modeling performance. As CNN is used as note encoder in both NSHNET and HAN, the main difference lies in the MoE Transformer of patient encoder, which demonstrates the importance of this model component. Moreover, NSHNET is the only text-based model that matches GRU-D in AUC-PRC, and it achieves the highest F1 score across the board.

C. Note Selection With NSHNET

Finally, we apply NSHNET to identify important input notes by calculating the attention flow values of different notes with respect to the [CLS] token. Notes with highest flow values are selected as input to fine-tune and evaluate BERT models. Table IV shows that note selection using NSHNET significantly improves the performance of three BERT variations. After the selection, we could achieve similar or better performance compared to other text-based baselines, but using only a fraction of total inputs (512 tokens). The results demonstrate the usefulness of NSHNET in assessing note contributions through attention flow. As recent work has shown that the BERT-based model can be used as all-purpose prediction engines for health systems [95], a tool like NSHNET can be useful to select the most informative notes to promote accuracy and efficiency in such a modeling setup.

V. DISCUSSION

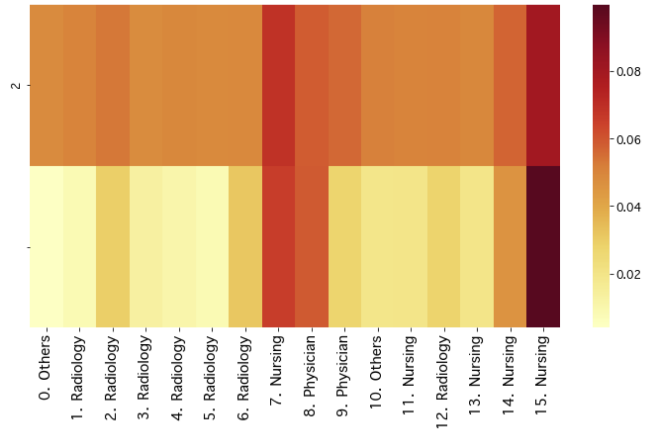
Clinical predictive analysis has many important applications in healthcare but is a challenging task due to the complexity of the clinical data. In this work, we investigate the performance of LLMs in predicting in-hospital mortality based on clinical notes, which in real world are diverse in style, complex in content,

and lengthy in form. We find that the majority of small-sized LLMs perform poorly on the task and only a few models could achieve an acceptable performance of AUC-ROC over 0.6, with many not being able to provide a numerical risk score. This changed after using larger models, which show significantly better performances while reducing the missing cases. The best model, LLAMA-3.3-70B-INSTRUCT, could obtain an AUC-ROC of 0.781 and AUC-PRC of 0.403 on the task. However, these remain largely below those of existing supervised models. The gap between these models with supervised baselines demonstrate the challenges of processing routinely collected clinical notes that are diverse and lengthy. This is further evidenced by the poor performance of the models when using advanced prompting strategies like CoT and SC, which in fact degrade the results, sometimes by a large margin. In addition, the medically adapted LLMs that are proficient in multiple-choice question answering, such as Llama-UltraMedical [70], also perform poorly, showing the unique challenges of modeling real-world clinical text. These outcomes aligns with a recent study [98] showing that LLMs are less effective for clinical prediction tasks, although it was based on clinical measurements rather than clinical text. Thus, our study highlights the importance of exercising caution when applying generative LLMs to high-stakes clinical decision-making, despite their remarkable performance in other applications and domains.

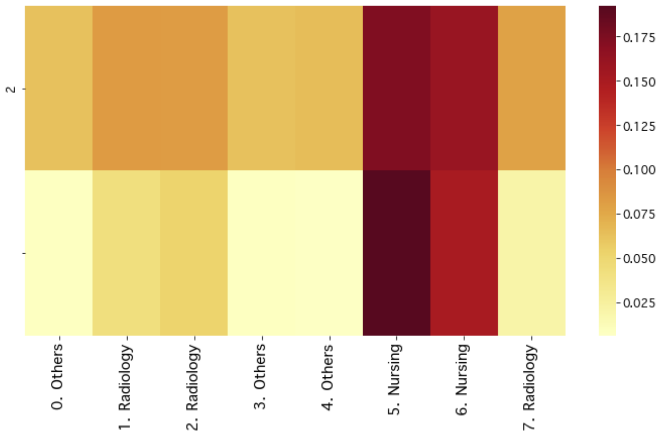
Many LLMs we examined would not present a valid risk assessment score in its generation. We found three scenarios where this happened. First, the model may directly refuse the request to perform risk assessment on the patient. Second, the model may only describe the patient case without giving any numerical value. Third, the input prompt may exceed the context length, where the models start to hallucinate or do not generate anything. We found significant numbers of the last two cases in the missing cases. Furthermore, even though some models have a long input context window, they sometimes still fail to provide a valid risk score. This could be related to the finding that long-context LLMs are not always effective in handling extended context [99], and the model performance could decrease as the input context increases [96]. Our poor results with few-shot prompting may be explained by these findings.

Fine-tuning decoder-based LLMs with LoRA proved highly effective, particularly when compared to common encoder-based models designed for fine-tuning. This underscores the advantages of larger models achieved through substantial pre-training budgets. Nevertheless, even with parameter-efficient fine-tuning (i.e., LoRA), the computational cost of training and deploying these models remains considerable, posing challenges for many healthcare systems [100]. In practice, we may opt to choose between using an out-of-the-box, computationally expensive model (i.e., large-sized LLMs for prompting) and customizing a cheaper, fit-for-purpose model (i.e., supervised models including small-sized LLMs with LoRA). Consequently, we focused on fine-tuning smaller LLMs in our experiments.

A hierarchical architecture can be an effective way to handle numerous clinical notes by maintaining the structure of the notes. We find that the proposed NSHNET model, which uses a hierarchical architecture with Transformer-based patient-level encoder, achieves better performance than all the text-based baselines in AUC-PRC and F1, including strong fine-tuned Transformer encoders and decoders. To our surprise, NSHNET even approaches or surpasses the strong measurement-based results, indicating the potential to further enhance text-based models for clinical risk prediction. By introducing the mechanism of attention flow, the Transformer-based patient encoder in NSHNET can assign *importance* to clinical notes, which can significantly reduce the input length. We show that this is beneficial for improving the performance of BERT models, but a future direction would be analyzing its impact on prompt-based LLMs.



(a) A true positive case (hospital admission ID: 140499)



(b) A false positive case (hospital admission ID: 170606)

Fig. 5. Two patient examples whose notes are weighed by the attention flow values in the patient encoder (a two layer Transformer). Darker colors mean higher attention flow values, thus more important notes.

Attention flow enjoys many benefits as a model explanation method since it is mathematically equivalent to Shapley Values [97], which is a more generalized and expressive method compared to SHAP [101]. Fig. 5 shows two examples to visualize the note contributions based on attention flow. NURSING notes are observed to be more important for the mortality prediction task, which is consistent with previous findings [46], [59]. Identifying important notes could provide interpretations to clinicians and allow them to read the original content to avoid unexpected modeling behaviors [44], [45]. For example, we examined a false positive prediction made by NSHNET in Fig. 5(b) and dived into the nursing notes tagged red (with high

attention flow values, thus highly important). It turned out that the notes described the patient to have multiple complications, including cancer, sepsis, and mental deterioration, who was later discharged to the hospice home and passed away within a week. In this case, the model actually makes a reasonable prediction to assess patient condition, although its prediction is false given the *in-hospital* mortality label.

VI. Limitations and Future Work

We examined a wide range of open-source LLMs, but the list is by no means exhaustive. However, we believe that they are representative of the current open source models. We also did not include proprietary models given the data usage agreement restrictions, which do not allow redistribution of the data to third parties, such as via an API to a commercial LLM [2]. However, certain models we used, especially Llama-3.3-70B-Instruct, have been shown to be on par or slightly better than *GPT-4-1106-preview* and *GPT-4-Turbo-2024-04-09* based on the Chatbot Arena leaderboard [102]. Hence we consider them reasonable choices for our study.

We observed harmful effects of few-shot examples for the prediction task, but this is confounded by the possible decreasing capabilities of LLMs to handle longer contexts. Future work needs to investigate how to effectively use few-shot examples in the clinical context, such as creating succinct summaries to replace the original text.

We primarily focused on in-hospital mortality for the prediction task. Meanwhile, as the case in Fig. 5(b) shows, the model may correctly predict patient risk beyond the duration of hospital stay. Future work may enrich the task setup by considering mortality in long time spans as well as other predictive tasks. Lastly, we used attention flow to select important notes, but we acknowledge that the current selection is task-specific. How to select informative notes in general requires a refined definition of note importance and likely an unsupervised experimental setup.

VII. Conclusion

In this study, we examined the text-based prediction of in-hospital mortality based on routinely collected clinical notes of diverse categories. We benchmarked 34 popular open-source LLMs and found that the large-size LLMs could only perform modestly on the task, with the best model – Llama-3.3-70B-Instruct – achieving an AUC-ROC below 0.8, substantially lower than supervised baselines. To advance the supervised performance of text-based modeling, we proposed a supervised, hierarchical architecture with specialized clinical note modules called NSHNet, which considers the variations in different clinical note categories. The model showed improved predictive performances compared to multiple baselines including LLMs, and could identify informative notes through attention flow. Our results show that supervised methods are still preferred on this prediction task for ICU patients, and that current LLMs lag behind despite of various prompting and decoding methods. Finally, we release the patient cohort, LLM inference script with prompts, and a reference implementation of NSHNet in https://github.com/JHLiu7/text-based-clinical-risk-predictions.

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